

Curriculum Vitae

Full name Manfred Buchacher
Date of birth 12/05/1989
Nationality Austrian
eMail manfredi.buchacher@gmail.com
Web page buchacm.github.io

Employment

01/2024 - 03/2026 **Project leader**, Johannes Kepler University Linz
project title: **Problems of separation in enumerative combinatorics**

11/2022 - 12/2023 **Postdoctoral researcher**, Austrian Academy of Sciences
mentor: Josef Schicho

12/2021 - 07/2022 **Postdoctoral researcher**, Johannes Kepler University Linz
mentor: Manuel Kauers

10/2017 - 11/2021 **Project assistant**, Johannes Kepler University Linz
project title: **Algorithmic lattice path counting using the kernel method**
supervisor: Manuel Kauers

12/2016 - 09/2017 **Project assistant**, Technical University Vienna
project title: **Communication and complexity constrained inference over graphs for big data**
supervisor: Gerald Matz

01/2016 - 09/2016 **Trainee**, UniCredit Bank Austria
Market Risk Management & Price Control

Education

10/2017 - 11/2021 **PhD in Mathematics**, Johannes Kepler University Linz,
thesis: **Algorithms for the enumeration of lattice walks**,
supervisor: Manuel Kauers

10/2012 - 04/2015 **Master of Science in Mathematics**, University of Vienna
thesis: **The asymptotic behaviour of denumerable markov chains**,
supervisor: Jiří Černý

03/2009 - 09/2012 **Bachelor of Science in Mathematics**, University of Vienna
thesis: **Ehrhart theory**
supervisor: Ilse Fischer

Research interests

algorithmic enumerative combinatorics, constructive algebra, effective algebraic geometry, differential algebra, dynamical systems

Awards and scholarships

- 01/2024 - 02/2026 **LIT Young Career Award**, LIT-2022-11-YOU-214, 179 591.67 EURO
08/2014 - 03/2015 **Scholarship for Master's Students**, FWF
2012/2013 **Merit Scholarship**, University of Vienna
2009/2010 **Merit Scholarship**, University of Vienna

Publications and preprints

- $x(1 - t(x + x^{-1}))F(x; t) = x - tF(0; t)$, 2025, arXiv:2512.21753
- *On finite orbits of infinite correspondences*, 2025, arXiv:2511.07172
- *Separated variables on plane algebraic curves*, 2024, arXiv:2411.08584
- *Separating variables in bivariate polynomial ideals: the local case*, 2024, arXiv:2404.10377
- *On the problem of separating variables in multivariate polynomial ideals*, with Manuel Kauers, Proceedings of ISSAC'24, 2024, DOI:10.1145/3666000.3669680
- *The Newton-Puiseux algorithm and effective algebraic series*, Journal of Algebra, 2026, DOI:10.1016/j.jalgebra.2025.09.027
- *The orbit-sum method for higher order equations*, with Manuel Kauers, 2022, to appear in Annals of Combinatorics, arXiv:2211.08175
- *Quadrant Walks Starting Outside the Quadrant*, with Manuel Kauers and Amélie Trotignon, Proceedings of FPSAC'21, 2021, Corpus ID: 227210243
- *Walks with Small Steps in the 4D-Orthant*, with Sophie Hofmanninger and Manuel Kauers, Annals of Combinatorics, 2020, DOI:10.1007/s00026-020-00520-5
- *Separating Variables in Bivariate Polynomial Ideals*, with Manuel Kauers and Gleb Pogudin, Proceedings of ISSAC'20, 2020, DOI:10.1145/3373207.3404028
- *Inhomogeneous Restricted Lattice Walks*, with Manuel Kauers, Proceedings of FPSAC'19, 2019, Corpus ID: 119580728

- *Graph Learning based on Total Variation Minimization*, with Peter Berger, Gábor Hannák and Gerald Matz, 2018 IEEE International Conference on Acoustics, Speech and Signal Processing, 2018, DOI:10.1109/ICASSP.2018.8461603

Selected presentations

- 11/2025 **AROMATH**, INRIA Centre at Université Côte d'Azur, France
- 09/2025 **ÖMG - DMV Annual Meeting**, Linz, Austria
- 07/2025 **Applications of Computer Algebra**, Heraklion, Greece
- 06/2025 **Number Theory Seminar**, Université Paul Sabatier, Toulouse, France
- 02/2025 **Enumerative Combinatorics and Effective Aspects of Differential Equations**, CIRM, Marseille, France
- 02/2025 **Galois Differential Theories and Transcendence**, CIRM, Marseille, France
- 01/2025 **Arbeitsgemeinschaft Diskrete Mathematik**, University of Vienna, Austria
- 11/2023 **SFB AEC Status Seminar**, Admont, Austria
- 07/2023 **Applications of Computer Algebra**, Warsaw, Poland
- 06/2023 **Algebraic and Tropical Methods for Solving Differential Equations**, Oaxaca, Mexico
- 12/2022 **SFB AEC Status Seminar**, Admont, Austria
- 01/2022 **Formal Power Series and Algebraic Combinatorics**, Ramat Gan, Israel, Online
- 11/2021 **SFB AEC Status Seminar**, Online
- 07/2021 **Applications of Computer Algebra**, Online
- 04/2021 **Séminaire de Calcul Formel**, Sorbonne Université and INRIA Saclay, France, Online
- 03/2021 **Groupe de travail Transcendence et Combinatoire**, l'Institut Henri Poincaré and INRIA Saclay, France, Online
- 12/2020 **SFB AEC Status Seminar**, Online
- 11/2019 **Laboratory of Theoretical Computer Science**, HSE University, Moscow, Russia
- 07/2019 **SIAM Algebraic Geometry**, Bern, Switzerland
- 07/2019 **Formal Power Series and Algebraic Combinatorics**, Ljubljana, Slovenia
- 02/2019 **Special Colloquium**, Université de Tours, France
- 12/2018 **SFB AEC Status Seminar**, Strobl am Wolfgangsee, Austria

Conferences, seminars, summer schools and workshops

- 11/2025 **AROMATH**, INRIA Centre at Université Côte d'Azur, France
- 09/2025 **ÖMG - DMV Annual Meeting**, Linz, Austria
- 07/2025 **Applications of Computer Algebra**, Heraklion, Greece
- 06/2025 **Number Theory Seminar**, Université Paul Sabatier, Toulouse, France
- 02/2025 **Enumerative combinatorics and effective aspects of differential equations**, CIRM, Mar
- 02/2025 **Galois differential theories and transcendence**, CIRM, Marseille, France
- 10/2024 **Algebraicity and Transcendence for Singular Differential Equations**,
Erwin Schrödinger Institute, Vienna, Austria
- 11/2023 **SFB Statusseminar**, Admont, Austria
- 06/2023 **Algebraic and Tropical Methods for Solving Differential Equations**, Oaxaca, Mexico
- 12/2022 **SFB Statusseminar**, Admont, Austria
- 07/2022 **Algorithmic and Enumerative Combinatorics**, Technical University Vienna, Austria
- 11/2021 **SFB Statusseminar**, Online
- 08/2021 **SFB in-person meeting**, JKU Linz, Austria
- 12/2020 **SFB-Statusseminar**, Online
- 01/2020 **SFB-Colloquium**, RISC, Hagenberg, Austria
- 12/2019 **SFB-Statusseminar**, BIFEB, Strobl am Wolfgangsee, Austria
- 10/2019 **SFB-Colloquium**, Technical University Vienna, Austria
- 08/2019 **AEC Summer School**, RISC, Hagenberg, Austria
- 06/2019 **15th International Symposium on Orthogonal Polynomials, Special Functions
and Applications**, RISC, Hagenberg, Austria
- 06/2019 **SFB-Colloquium**, RICAM, Linz, Austria
- 04/2019 **SFB-Colloquium**, University of Vienna, Austria
- 01/2019 **SFB-Colloquium**, RISC, Hagenberg, Austria
- 12/2018 **SFB-Statusseminar**, BIFEB, Strobl am Wolfgangsee, Austria
- 10/2018 **SFB-Colloquium**, Technical University Vienna, Austria
- 08/2018 **AEC Summer School**, RISC, Hagenberg, Austria
- 06/2018 **SFB-Colloquium**, JKU Linz, Austria
- 05/2018 **Combinatorics, Special Functions and Computer Algebra – A workshop on the
occasion of Peter Paule's 60th birthday**, RISC, Hagenberg, Austria
- 03/2018 **SFB-Colloquium**, University of Vienna, Austria
- 01/2018 **SFB-Colloquium**, RISC, Hagenberg, Austria

- 12/2017 **SFB-Statusseminar**, BIFEB, Strobl am Wolfgangsee, Austria
 11/2017 **Algorithmic and Enumerative Combinatorics**, Erwin Schrödinger
 Institute, Vienna, Austria

Teaching

Johannes Kepler University Linz

- Spring 2024 **Algebra für InformatikerInnen**, teaching assistant
 Spring 2021 **Algebra für InformatikerInnen**, teaching assistant
 Fall 2020 **Diskrete Strukturen**, teaching assistant
 Spring 2020 **Algebra für InformatikerInnen**, teaching assistant
 09/02 - 13/02/2020 Projektwoche Angewandte Mathematik: **Abzählende Kombinatorik und das Zählen von Jongliermustern**, seminar for high school students
 Fall 2019 **Diskrete Strukturen**, teaching assistant
 Spring 2019 **Algebra für InformatikerInnen**, teaching assistant
 01/02/2019 **Mathematik und Zauberei: Wie mathematische Ideen die Grundlage von Zaubertricks bilden**, seminar for high school students
 Fall 2018 **Stochastik für Lehramt**, teaching assistant
 Spring 2018 **Algebra für InformatikerInnen**, teaching assistant

University of Vienna

- Fall 2013 **Tutorium für LehramtskandidatInnen zur STEOP**, teaching assistant

Co-supervision

- Sophie Hofmanninger *4-dimensional lattice walks restricted to the positive orthant*
 Master's thesis, Johannes Kepler University Linz, 2020

Refereeing

Combinatorial Theory, FPSAC, ISSAC, Journal of Algebraic Combinatorics, Journal of Symbolic Computation, La Matematica, Séminaire Lotharingien de Combinatoire

Languages

German (native), English (fluent), French (intermediate)

Software

Latex, Mathematica, Maple, SageMath, Matlab, Java, Microsoft Excel, Visual Basic